Topic B: proof routes and acquisition costs

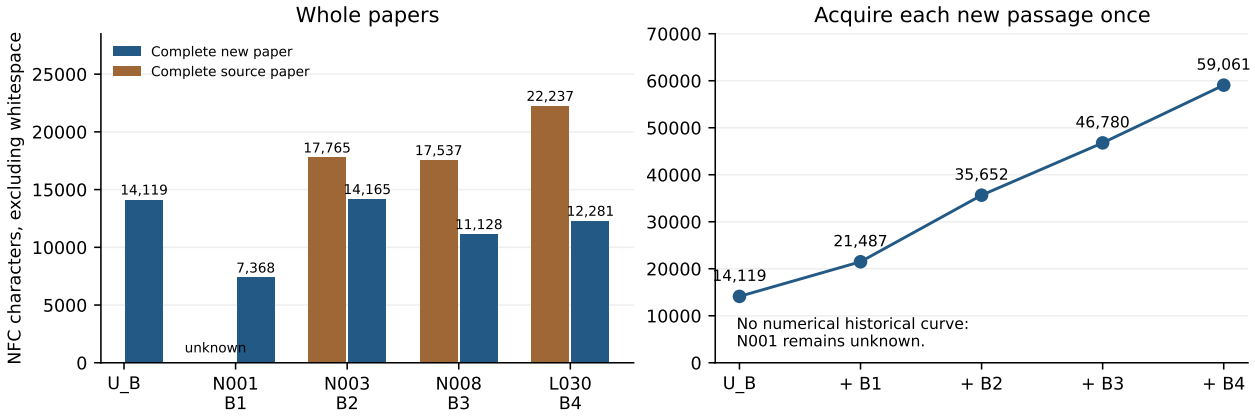
Human Compression project

The four fixed contracts cover thirty clauses from Buekenhout–Cara (1998), Cameron–Cara (2002), Archer–Cara–Krempa (2005), and the revised Garcia-Colin–Leemans manuscript. The common theory is *Forest coordinates and relative generation*. Its four complete applications retain the admitted hypotheses, outside theorems, corrections and exclusions.

The complete comparison. The whole common paper costs 14,119 characters; the four applications cost 44,942. Three source PDFs contribute 57,539. If x is the unknown original PDF cost of N001, the complete comparison is

$$R_B = \frac{x + 57\,539}{59\,061}.$$

There is no numerical value of R_B. The N001 application remains fully charged. Its authored browser text supplies documentary evidence but is not substituted for the missing PDF measurement.



The unit throughout this page is NFC characters without whitespace in the complete extracted PDF text, using the reviewed furniture exclusions and glyph corrections. Image and expanded outside-proof costs remain unknown.

After acquiring the whole common theory. The following admitted factors divide the complete source paper by its complete application. They do not discount the umbrella in the overall comparison or charge it again at each application.

Source / application	Source	Application	Conditional factor
N001 / B1	unknown	7,368	unknown
N003 / B2	17,765	14,165	1.254
N008 / B3	17,537	11,128	1.576
L030 / B4	22,237	12,281	1.811

This proof edition has no required mathematical program dependency. Validation infrastructure and the six scoped Lean declarations are separately inventoried; they are not claims of complete formalization.

One explicit acquisition order. Start with the complete U_B, then acquire B1, B2, B3 and B4. Let K_i be the union of passage owners paid through stage i. The added cost is $c(K_i \setminus K_{i-1})$; every complete local paper, including context and references, belongs to K_i. Recorded prerequisites already in K_{i-1} are not charged again. The table is an exact union calculation, not a proposed shortest proof or an estimate of time spent.

Paper added	New cost	Cumulative	Prior passages used
U_B	14,119	14,119	—
B1	7,368	21,487	6,995
B2	14,165	35,652	10,208
B3	11,128	46,780	6,450
B4	12,281	59,061	5,314

The last column counts the already paid passages appearing in that application's recorded proof route. It is not an extra subtraction from the new cost. Every historical cumulative stage is symbolic: x, then $x + 17\,765$, $x + 35\,302$, and $x + 57\,539$.

What the paired graph explains. The offline reader selects a whole contract or any of its thirty clauses. Arrows run from prerequisite to use. It offers a compact named-result view and an expanded proof-branch view. Selecting a box gives the precise source or current proof location, its paid passages and retained assumptions. Comparison, attribution and verification links are displayed separately from proof prerequisites. The five repaired documentary interfaces in the accepted logical revision remain in force, including the implicit Kneser connectedness premise and the two Whiston inputs' retained classification provenance.

The graph contains 73 historical nodes, 85 new nodes, 19 retained external inputs and seven verification components: 184 nodes and 348 fine edges. These categories include definitions, statements and proof branches. They are not comparable counts of results, and no node-count factor is reported. There are 217 disjoint spans with 214 text owners. Parent boxes and proof branches can share owners: box costs overlap, while totals count each passage once. Statement-only source passages are labeled as such; a proof charged elsewhere does not become an additional cost here.

Result routes and whole papers answer different questions. A selected prerequisite union explains which recorded arguments support a result. It can omit context that is still paid in the complete-paper total. Neither this union nor a graph traversal is a minimum proof cost. The full umbrella remains paid within Topic B, independently of other topics. External proofs, missing original-PDF text and unextracted graphics remain unknown in both views.

Reproduction and scope. The standalone HTML retains all thirty clause mappings and all fine edges; its JSON files bind the reviewed graph, corrected ownership layer and admitted measurements. The generated figures and this reader are derived explanations. They do not change any mathematical contract, source credit, question count or previously admitted factor. A separate checker verifies complete owner unions, input hashes and the absence of an invented historical curve; the UI harness tests all contract and clause routes. No primary-source text or images are distributed in this reader.